Future Capabilities Roadmap

This file lists future capabilities that fit the current codebase structure. The current codebase now includes a GUI, config-driven multihit runner, action-file creator, post-processing tools, DOE orchestration, dataset creation, Bayesian proposal, and visualization/reporting. The items below can be added without replacing that architecture.

Run setup and robustness

- Preflight validation in the GUI and runner that checks all key paths, action columns, material names, DEFORM object names, die axis settings, stock diameter, and output folders before any DB is written.
- Restart/resume support that reads the manifest and skips completed hits or completed candidate runs.
- Stronger failure classification, such as DEFORM abnormal stop, missing DB, post-processing failure, invalid geometry, or missing output file.
- Automatic run archiving that stores the exact config, action CSV, key file name, code version, and environment notes with every run.

Geometry and positioning

- Coordinate-driven die recentering before every hit, using Workpiece, Top Die, and Bottom Die center coordinates.
- Optional bending and straightness metrics from workpiece centerline or surface point tracking.
- Automatic detection of whether the configured die axis should be Z or Y based on key-file object coordinates.
- More general multi-die support where each die pair has named object aliases and axis conventions in the config.
- Tooling to compare final workpiece geometry across DOE candidates.

Material and microstructure

- Material library discovery that lists available DEFORM material names directly from the system and user material folders, not only from `material_list_deform.csv`.
- Material sweeps where candidate rows can select material source, material name, or material number.
- Supplier grain-size initialization from distributions, not only a single mean grain size.
- Lag-time modeling between hits so JMAK grain growth can occur during hold periods or transfer delays.
- JMAK calibration workflows that fit model parameters to measured grain-size data.
- Microstructure target matching that compares predicted grain statistics against experimental measurements by location or region.

Process variables and DOE

- Candidate support for speed, depth, temperature, height, time step, friction, die axis, die motion mode, lag time, stock diameter, and material choice.
- Per-hit candidate control where different hits can receive different depths, speeds, temperatures, or offsets.
- DOE templates for common studies, such as small single-die sweeps, both-dies-moving anti-bending sweeps, material comparisons, and JMAK calibration studies.
- Batch sizing rules that limit total runs, total estimated wall time, or total DEFORM solves per round.

Optimization

- Bayesian optimization with constraints, such as maximum damage limits, target grain-size ranges, or maximum load limits.
- Multi-objective optimization that tracks tradeoffs between damage, grain size, load, cycle time, and dimensional accuracy.
- Active learning that proposes candidates where the surrogate uncertainty is high.
- Trust-region optimization that proposes conservative changes near successful runs.
- Later reinforcement-learning experiments once the dataset and simulator interface are stable enough.

Post-processing and reporting

- Unified run reports that combine load-stroke curves, coordinate histories, damage summaries, microstructure summaries, and key config values.
- Live DOE dashboards that update as the manifest changes.
- Automatic comparison reports between baseline, symmetric-die, and single-top-die runs.
- Export packages for collaborators, including config, action CSV, summary metrics, plots, and notes.

Data management

- Stable dataset schemas with version tags so old DOE results remain readable after the code evolves.
- A run registry that maps every dataset row back to its config, action CSV, run folder, processed outputs, and original candidate row.
- Data-quality checks that flag missing microstructure, missing load-stroke data, incomplete hit folders, or inconsistent manifest status.
- Optional storage of large output files outside OneDrive while keeping metadata and summaries in the project folder.

User experience

- A stronger guided wizard inside the GUI for first-time users that explains the minimum choices needed for a run.
- GUI preflight warnings for common mistakes such as missing `_jmak.key`, action CSV version not found, or hit folders not found under `AF Data`.
- Config templates for common cases: default single-top-die, symmetric BDM, Y-axis dies, material sweep, and DOE optimization.
- A richer manual browser that opens specific PDFs from the GUI rather than only the manual folder.
- Clearer SOPs for adding a new material, adding a new die, starting a DOE, processing results, and proposing next candidates.